

Ian Laver is chasing his chickens and they act crazy! His velocity at any time t (in seconds after the chicken escaped) is modelled by the function $\sin(t^2)$. Construct a function will give you the total distance traveled x^2 seconds since the chickens escaped?

Find the derivative of the function you find above

Let $F(x) = \int_0^{\sin x} \frac{1}{1-t^2} dt$. Find the value of $F'(\pi)$.

A honeybee population starts with 100 bees and increases at a rate of $n'(t)$ bees per week. What does $100 + \int_0^{15} n'(t) dt$ represent?

- A. The rate at which the bee population increases after 15 weeks have passed.
- B. The total bee population after 15 weeks.
- C. The average rate at which the bee population increases over the first 15 weeks.
- D. The total increase in the bee population after 15 weeks.

4. (2 marks) The integral $\int_2^1 (\sin(x^2) - 3) \, dx$ is

(A) negative.

(B) zero.

(C) positive.

5. (2 marks) Let $f(x)$ be the amount of fuel in litres spent per minute to travel a road with slope x .

Let $x(t)$ be the slope of the road t minutes into the trip.

Which of the following gives the total amount of fuel spent travelling in the first 5 minutes?

(A) $\int_0^5 f(x(t)) \, dt$

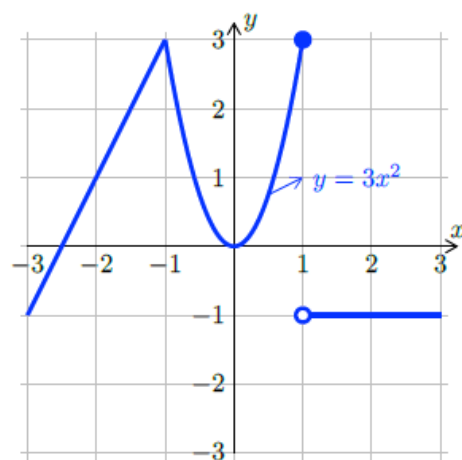
(B) $\int_0^5 f'(x(t)) \, dt$

(C) $\int_{x(0)}^{x(5)} f(x) \, dx$

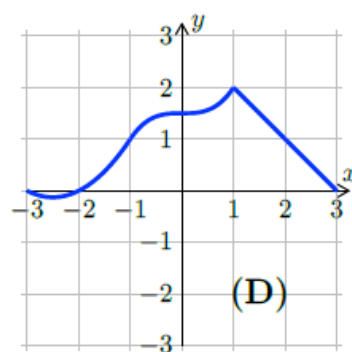
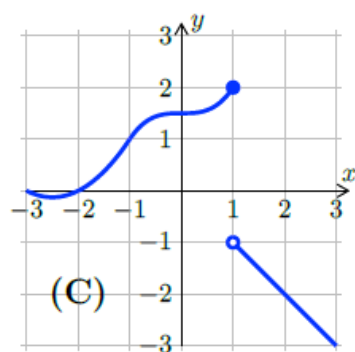
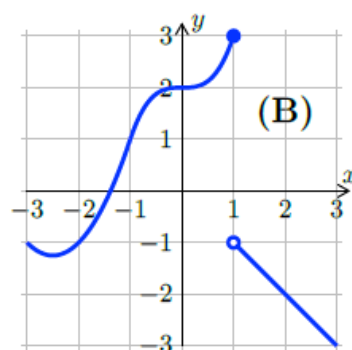
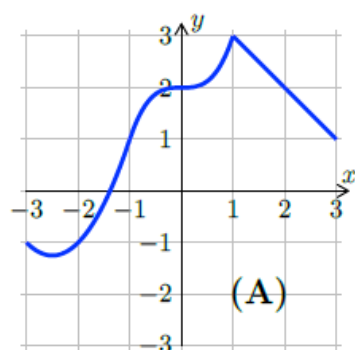
(D) $\int_{x(0)}^{x(5)} f(x(t)) \cdot x'(t) \, dt$

(E) $\frac{1}{5} \int_0^5 f(x(t)) \, dt$

6. (2 marks) Consider the function $y = f(x)$ given by the graph below.



Which of the following graphs corresponds to its continuous antiderivative that passes through the point $(-1, 1)$?



7. (2 marks) According to <https://ourworldindata.org/co2-emissions>, the rate at which CO₂ is produced in the world by burning fossil fuels and cement production alone can be approximated by

$$r(t) = 9.35 \cdot e^{0.03t}$$

measured in millions of tons, where t is the number of years since 1750.

Which of the following functions measures the total amount of CO₂ produced since 1900 by burning fossil fuels and cement production?

(A) $\int_0^t 9.35 \cdot e^{0.03x} dx$

(B) $\int_{1900}^t 9.35 \cdot e^{0.03x} dx$

(C) $\frac{9.35}{0.03} e^{0.03t}$

(D) $\frac{9.35}{0.03} (e^{0.03t} - e^{4.5})$

8. (2 marks) Consider the differential equation

$$\frac{dy}{dx} = -2xe^{x^4}.$$

Which of the following functions is a solution of this differential equation?

(A) $y = \int_1^{-x^2} e^{-t^2} dt$

(C) $y = 2 - \int_1^{-2x} e^{t^4} dt$

(B) $y = 1 + \int_{2x}^1 e^{t^4} dt$

(D) $y = 3 + \int_{x^2}^1 e^{t^2} dt$

2 Consider the following differential equation: (4 marks)

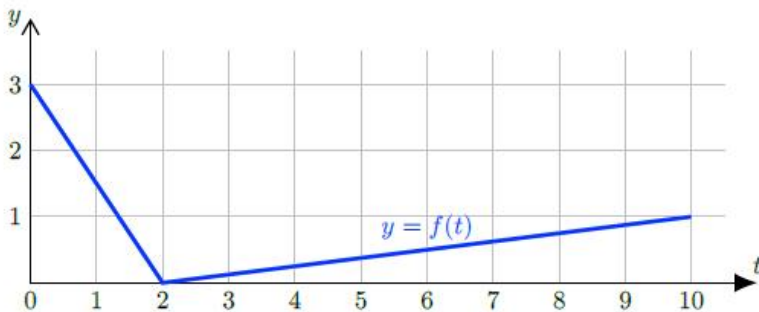
$$\frac{dy}{dx} = -3x^2 e^{-x^6}$$

Which of the following functions is a solution of this differential equation?

Fill in exactly one option for this question.

- (A) $f(x) = 1 + \int_0^{x^3} e^{-t^2} dt$
- (B) $g(x) = 2 + \int_{x^3}^0 e^{-t^2} dt$
- (C) $h(x) = 3 + \int_0^{x^3} e^{-x^2} dx$
- (D) $k(x) = 4 + \int_{x^3}^0 e^{-x^2} dx$

6 (7 marks) We want to approximate the signed area under the graph below.



To get a good estimate, we will use 100 intervals.

What kind of Riemann sums will give the best approximation?

Fill in exactly one option for this question.

- (A) Left-hand Riemann sum: $\sum_{i=1}^{100} f(t_{i-1}) \Delta t$
- (B) Midpoint Riemann sum: $\sum_{i=1}^{100} f\left(\frac{t_{i-1} + t_i}{2}\right) \Delta t$
- (C) Right-hand Riemann sum: $\sum_{i=1}^{100} f(t_i) \Delta t$
- (D) All three approximations above are equally accurate.

8 Hummingbirds are known for hovering in the air while flapping their wings at (32 marks)

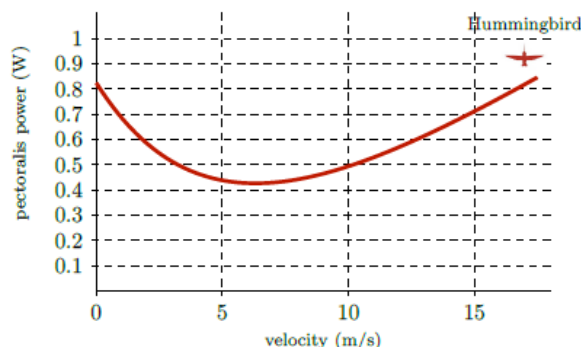
incredible speeds. They are however not the birds that spend the most energy to fly. In a Science article¹, the authors analyze the power that hummingbirds require to fly and compare it to other birds.

The figure on the right is adapted from the paper.

It shows the power (in Watts W) spent by a hummingbird as a function of their flight speed.

We want to measure the energy that a hummingbird spends in the following 7-second flight:

t (in seconds)	0	1	2	3	4	5	6
v (in m/s)	0	5	5	10	15	5	0



Recall that

$$\text{Energy} = \int (\text{Power}) \, dt,$$

so the units of Energy are Joules (J) and $1 \text{ J} = 1 \text{ W}\cdot\text{s}$ (Watts times seconds).

(a) Consider the function: (6 marks)

- $P(t)$ = Power needed by the hummingbird t seconds into the flight (in W).

Estimate $P(0)$, $P(1)$, and $P(2)$.

$$P(0) \approx \boxed{} \text{ W} \quad , \quad P(1) \approx \boxed{} \text{ W} \quad , \quad P(2) \approx \boxed{} \text{ W}$$

(b) Estimate $P'(0)$ and $P''(0)$. Justify your answer. (8 marks)

$$P'(0) \approx \boxed{} \text{ W/s}$$

$$P''(0) \approx \boxed{} \text{ W/s}^2$$

This question, along with the following one, is a two-part problem in which we use real-life data to obtain a degree two Taylor polynomial at $t = 3$ of $f(t)$ and use it to approximate $f(6)$.

The following table shows the 7-day average of new daily COVID-19 cases in Ontario in the first few days of 2021

Date	01/01/2021	02/01/2021	03/01/2021	04/01/2021	05/01/2021
7-day ave	2480.86	2655.29	2792.29	2982.43	3064.57

Here t denotes the time (in days) since 31/12/2020 (so that the table above gives values between $f(1)$ and $f(5)$). Using the data from the table, estimate the right-sided derivative of $f(t)$ for $t = 3$ and $t = 4$, and use those two values to estimate the right-sided second derivative of $f(t)$ at $t = 3$. With estimates, construct a degree of two Taylor polynomial for $f(t)$ about $t = 3$ and use it to estimate the value of $f(6)$ with two decimal places.

Estimate the integral $\int_{-10}^2 \frac{x^2}{4} dx$ using a right-hand Riemann sum and $n = 4$ subdivisions.

☐ 52.5

☐ 44

☐ 127.5

☐ 124.5

Since 1987, the men's 100m world record has never stood for more than three years and three months. Until, that is, Jamaican legend Usain Bolt set the current world record in August 2009 - almost 13 years ago.

Usain Bolt set the current 100m world record at the 2009 IAAF World Championships, clocking an incredible 9.58 seconds for the feat. At the record-winning event in 2009, Bolt's average ground speed was 37.58km/h, whilst reaching a top speed of 44.72km/h in the 60-80m stretch – numbers fitting for the world's fastest man.

(excerpts from [source #1](#) and [source #2](#))

Let $B(t)$ be the velocity of Usain Bolt in his record setting 100m race (in km/h) t seconds after the start of the race.

Consider the following quantity

$$\max_{t \in [0, 9.58]} \frac{1}{t} \int_0^t B(s) \, ds.$$

Note: Recall that $\max_{t \in [a, b]} f(t)$ means the maximum value of $f(t)$ for all values of t between a and b .

(a) Select the best practical description for this quantity from the options below.

☐

(b) What are the units of this quantity?

☐

(c) From the information in the excerpt above, what can we say about this quantity?

☐

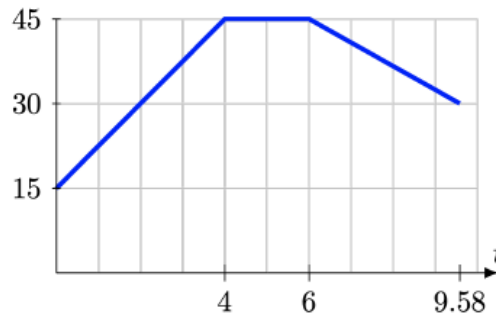
If we look more carefully at the previous question:

$$\min_{t \in [0, 9.58]} \frac{1}{t} \int_0^t B(s) \, ds,$$

we may notice that something "strange" is happening when t is close to 0.

Let us investigate what is happening at $t = 0$.

(a) For simplicity, let us assume that Usain Bolt actually reached a velocity of 45km/h and that his velocity function $B(t)$ had the following graph



[Graph Description: It is linear from (0,15) to (4,45), then it is constant from (4,45) to (6,45), then it is linear from (6,45) to (9.58, 30).]

We are focusing on values of t near $t = 0$, so if $0 \leq t \leq 4$, we have

$$\int_0^t B(s) \, ds = \square$$

You cannot use Riemann sums in this question.

(b) Divide the result by t and calculate

$$\lim_{t \rightarrow 0^+} \frac{1}{t} \int_0^t B(s) \, ds = \square$$

So even though the function we were minimizing seems to not be defined when $t = 0$, it actually has a finite limit.

(c) Let's look at the general case now. Consider a function

$$F(t) = \int_0^t B(s) \, ds.$$

Then $F(0) = \square$

And

$$\lim_{t \rightarrow 0} \frac{F(t)}{t} = \square = \square$$

In January 2019, the Journal of the Royal Society Interface reported on new research on the hagfish, an underwater creature that dwells in the ocean. To protect itself from sharks, a hagfish is able to release a cloud of slime that is many times the volume of the hagfish itself. As the Journal reported, the [slime] is so thick and fibrous, predators have little choice but to spit out the hagfish and try to clear their mouths. Researchers modelled the release of the slime. Consider the following two functions from a particular observation.

- Let $s(t)$ be the rate at which the hagfish is releasing the slime, measured in micrometers per second, t seconds after it starts to release the slime
- Let $v(x)$ be the rate at which the hagfish is releasing the slime, measured in micrometers per second, when the shark is x meters away from the hagfish

7 (2 marks) Which of the following expressions gives the exact total amount of slime the hagfish releases in the first z seconds of its slime release?

(A) $\int_0^z v(x) \, dx$

(B) $\frac{1}{z} \int_0^z v(x) \, dx$

(C) $\int_0^z s(t) \, dt$

(D) $\int_{v^{-1}(0)}^{v^{-1}(z)} s(t) \, dt$

(E) $\int_{v^{-1}(0)}^{v^{-1}(z)} v(x) \, dx$

8 (2 marks) Select the units of the integral

$$\int_a^b s^{-1}(v(y)) \, dy.$$

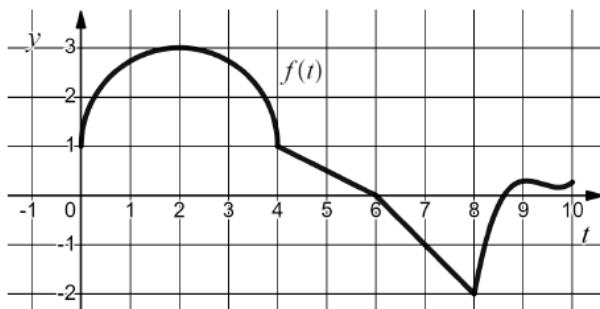
- (A) meters·micrometres
- (B) seconds·micrometres
- (C) seconds·metres
- (D) (metres·micrometres)/second
- (E) micrometres/second

6. Let f be a nonnegative continuous function on $[a, b]$. Let $c > 0$. Define a new function g by

$$g(x) = f(cx), \quad \text{where } \frac{a}{c} \leq x \leq \frac{b}{c}.$$

Let A denote the area under the graph of f on $[a, b]$, and let B denote the area under the graph of g on the interval $\left[\frac{a}{c}, \frac{b}{c}\right]$. Which of the following are true?

- (a) $A = B$
 - (b) $B = cA$
 - (c) $A = cB$
 - (d) $A = c^2B$
 - (e) $B = c^2A$
3. The function $y = f(t)$ is a piecewise defined function given in the following graph, which consists of a semicircle, two line segments and some unknown expression:



- (a) (3 marks) Compute $\int_0^4 (4f(t) - 3) dt$.

- (b) (3 marks) Compute $\int_4^8 |6f(t)| dt$.

- (c) (3 marks) Now consider the function $g(x) = \int_2^x f(t) dt$ where $f(t)$ is given in the graph above. Arrange the following according to size and briefly explain your reasoning: $g(2)$, $g(4)$, $g(6)$ and $g(9)$.

Answer:

$$g(\text{---}) \leq g(\text{---}) \leq g(\text{---}) \leq g(\text{---})$$

because: